ECET 250E

Linear Circuits 1

Week 5 | Chapter 6: Capacitors and Inductors

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Based on Fundamentals of Electric Circuits, 2026 Release — C.K. Alexander, M.N.O. Sadiku & S.J. Fredette (McGraw Hill)

Overview

So far we have limited our study to resistive circuits.

In this chapter, we shall introduce two new and important passive linear circuit elements:

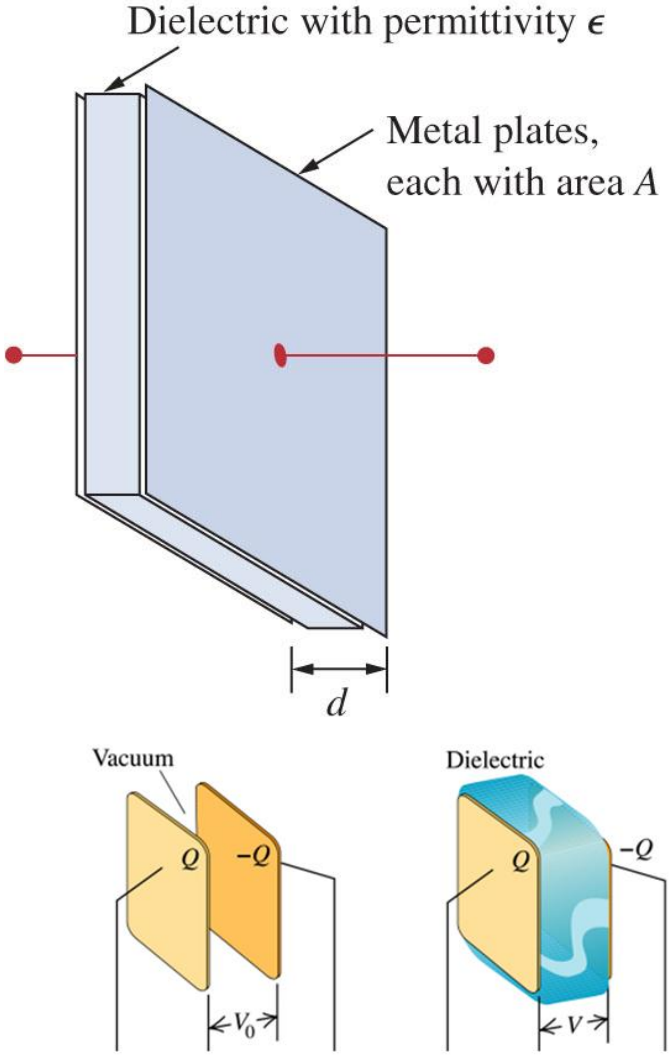
- The capacitor.
- The inductor.
- Unlike resistors, these elements do not dissipate energy.
- They instead store energy.
- We will also look at how to analyze them in a circuit.

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Capacitors [1]

- A capacitor is a passive element that stores energy in its electric field.
- It consists of two conducting plates separated by an insulator (or dielectric).
- The plates are typically aluminum foil.
- The dielectric is often air, ceramic, paper, plastic, or mica.

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Capacitors 2

- When a voltage source v is connected to the capacitor, the source deposits a positive charge q on one plate and a negative charge $-q$ on the other.
- The charges will be equal in magnitude.
- The amount of charge is proportional to the voltage:

$$q = Cv$$

- Where C is the capacitance.

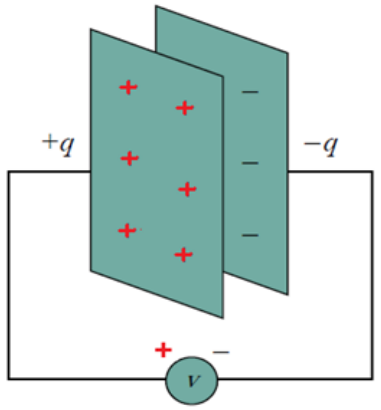


Figure 6.2 A capacitor with applied voltage v.

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Capacitors 3

The unit of capacitance is the Farad (F)

One Farad is 1 Coulomb/Volt

Most capacitors are rated in picofarad (pF) and microfarad (μF)

Capacitance is determined by the geometery of the capacitor:

- Proportional to the area of the plates (A).
- Inversely proportional to the space between them (d).
- ε is the permittivity of the dielectric.

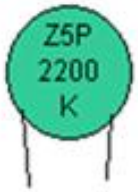
$$C = \frac{\varepsilon A}{d}$$

Three factors determine the value of the capacitance:

1. The surface area of the plates—the larger the area, the greater the capacitance.
2. The spacing between the plates—the smaller the spacing, the greater the capacitance.
3. The permittivity of the material—the higher the permittivity, the greater the capacitance.

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Reading capacitance values.



2200 pF ±10%

If a capacitor is marked "47," its value is 47 pf.

If it is marked .047, its value is .047 µf.

Thus, whole numbers express capacitance value in picofarads while decimal fractions express values in microfarads.

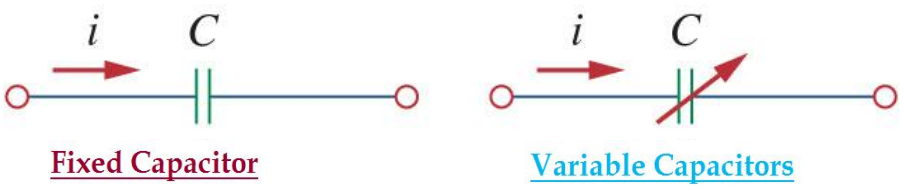
Any capacitor manufactured with a value of 1 µf or greater is physically large enough to be clearly marked with its actual value.

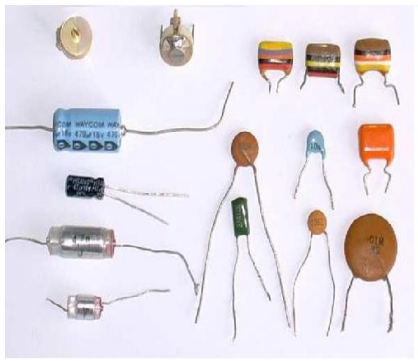
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Types of Capacitors

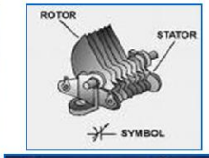
- The most common types of capacitors are film capacitors with polyester, polystyrene, or mica.
- To save space, these are often rolled up before being housed in metal or plastic films.
- Electrolytic caps produce a very high capacitance.
- Trimmer caps have a range of values that they can be set to.
- Variable air caps can be adjusted by turning a shaft attached to a set of moveable plates.

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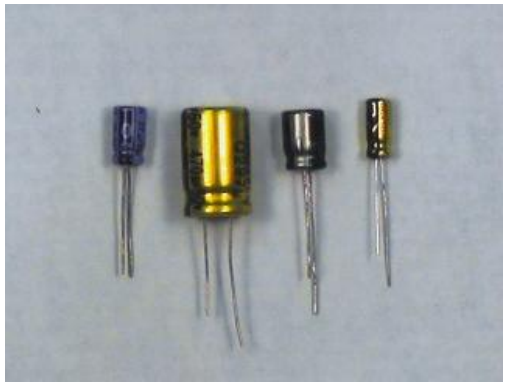






Electrolytic Capacitors





Polarity must be observed

Ceramic Capacitors





Polar Capacitor



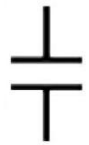


For the same capacitance value, polar capacitor are smaller in size, than non-polar capacitors.

Correct polarity must be maintained when connecting polar capacitors.

Non-polar Capacitor





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Applications for Capacitors

Capacitors have a wide range of applications, some of which are:

- Blocking DC.
- Passing AC.
- Shift phase.
- Store energy.
- Suppress noise.
- Start motors.

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(a) (b) (c)

Courtesy of Tech America

Current Voltage Relationship

- Using the formula for the charge stored in a capacitor, we can find the current voltage relationship.
- Take the first derivative with respect to time gives:

$$i = C\frac{dv}{dt}$$

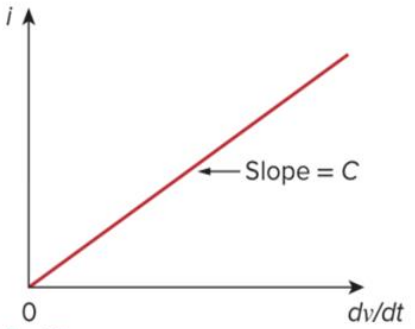


Figure 6.6
Current-voltage relationship of a capacitor.

- This assumes the passive sign convention.

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Since $$i = \frac{dq}{dt}$$

and $$q = Cv$$

$$\therefore \quad i = C\frac{dv}{dt} \tag{6.4}$$

The voltage-current relation of the capacitor can be obtained by integrating both sides of Eq. (6.4). We get

$$dv = \frac{1}{C}\, i\, dt$$

$$v = \frac{1}{C}\int_{-\infty}^{t} i\, dt \tag{6.5}$$

or $$\boxed{v = \frac{1}{C}\int_{t_0}^{t} i\, dt + v(t_0)} \tag{6.6}$$

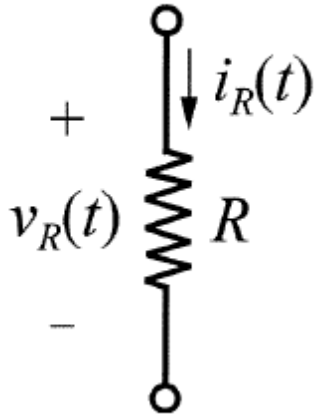
constant of integration

where $v(t_0)$ is the voltage across the capacitor at time t_0.

Equation (6.6) shows that capacitor voltage depends on the past history of the capacitor current.

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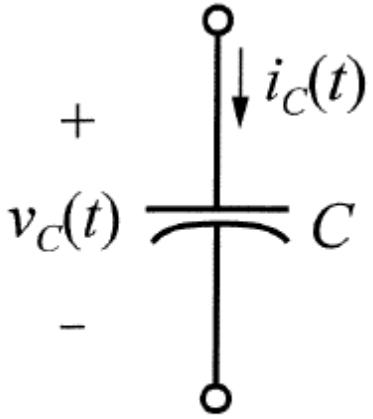
Resistor



$$v_R(t) = i_R(t)R$$

$$i_R(t) = \frac{v_R(t)}{R}$$

Capacitor



$$v_C(t) = \frac{1}{C}\int_{t_o}^{t} i_C(t)\,dt + v_C(t_o)$$

$$i_C(t) = C\frac{dv_C(t)}{dt}$$

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Stored Charge

- Similarly, the voltage current relationship is:

$$v(t) = \frac{1}{C}\int_{t_0}^{t} i(\tau)\,d\tau + v(t_0)$$

- This shows the capacitor has a memory, which is often exploited in circuits.
- The instantaneous power delivered to the capacitor is.

$$p = vi = Cv\frac{dv}{dt}$$

- The energy stored in a capacitor is:

$$w = \frac{1}{2}Cv^2$$

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The instantaneous power delivered to the capacitor is

$$p = vi = Cv\frac{dv}{dt} \tag{6.7}$$

The energy stored in the capacitor is therefore

[Since, we are now integrating w.r.t v, we must change the limit of integration to the intial voltage and final voltage.]

$$w = \int_{-\infty}^{t} p\,dt = C\int_{-\infty}^{t} v\frac{dv}{dt}dt = C\int_{v(-\infty)}^{v(t)} v\,dv = \frac{1}{2}Cv^2\Big|_{v(-\infty)}^{v(t)} \tag{6.8}$$

We note that $v(-\infty) = 0$, because the capacitor was uncharged at $t = -\infty$. Thus,

$$\boxed{w = \frac{1}{2}Cv^2(t)}$$

or simply

$$\boxed{w = \frac{1}{2}Cv^2} \tag{6.9}$$

Using Eq. (6.1), we may rewrite Eq. (6.9) as

$$w = \frac{q^2}{2C} \tag{6.10}$$

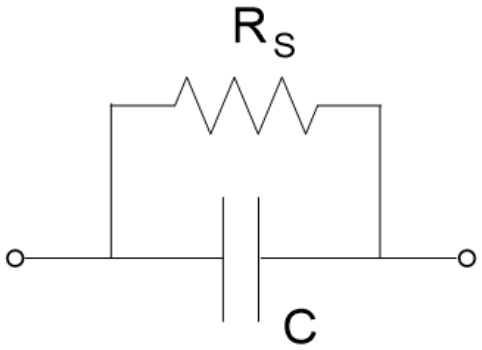
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Capacitor Properties

The capacitor has the following important properties:

3. The ideal capacitor does not dissipate energy. It takes power from the circuit when storing energy and returns previously stored energy when delivering power to the circuit.

4. A real, non-ideal, capacitor has a "leakage resistance" which is modeled as shown below. The leakage resistance may be as high as 100MΩ, and can be neglected for most practical applications.



Note: In this course we will always assume that the capacitors are ideal.

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Capacitor Properties

The capacitor has the following important properties:

1. When the voltage across a capacitor is constant (not changing with time) the current through the capacitor:

$$i = C\,dv/dt = 0$$

Thus, **a capacitor is an open circuit to dc.**

If, however, a dc voltage is suddenly connected across a capacitor, the capacitor begins to charge (store energy).

2. The voltage across a capacitor must be continuous, since a jump (a discontinuity) change in the voltage would require an infinite current, which is physically impossible. Thus, a capacitor resists an abrupt change in the voltage across it, Conversely

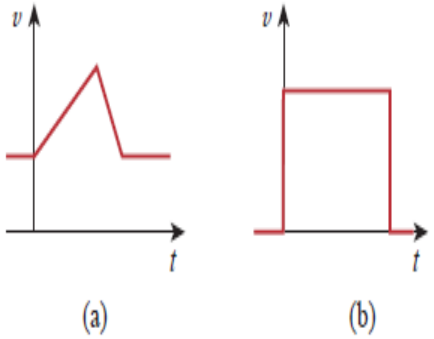


Figure 6.7
Voltage across a capacitor: (a) allowed, (b) not allowable; an abrupt change is not possible.

The voltage across a capacitor cannot change instantaneously

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Properties of Capacitors 1

- Ideal capacitors all have these characteristics:
- When the voltage is not changing, the current through the cap is zero.
- This means that with DC applied to the terminals no current will flow.
- Except, the voltage on the capacitor's plates can't change instantaneously.
- An abrupt change in voltage would require an infinite current!
- This means if the voltage on the cap does not equal the applied voltage, charge will flow and the voltage will finally reach the applied voltage.

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Properties of Capacitors 2

- An ideal capacitor does not dissipate energy, meaning stored energy may be retrieved later.
- A real capacitor has a parallel-model leakage resistance, leading to a slow loss of the stored energy internally.
- This resistance is typically very high, on the order of 100 MΩ and thus can be ignored for many circuit applications.

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EXAMPLE 6.1

(a) Calculate the charge stored on a 3-pF capacitor with 20 V across it.
(b) Find the energy stored in the capacitor.

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EXAMPLE 6.2

The voltage across a 5-μF capacitor is

$$v(t) = 10\cos 6000t \text{ V}$$

Calculate the current through it.

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EXAMPLE 6.3

Determine the voltage across a 2-μF capacitor if the current through it is

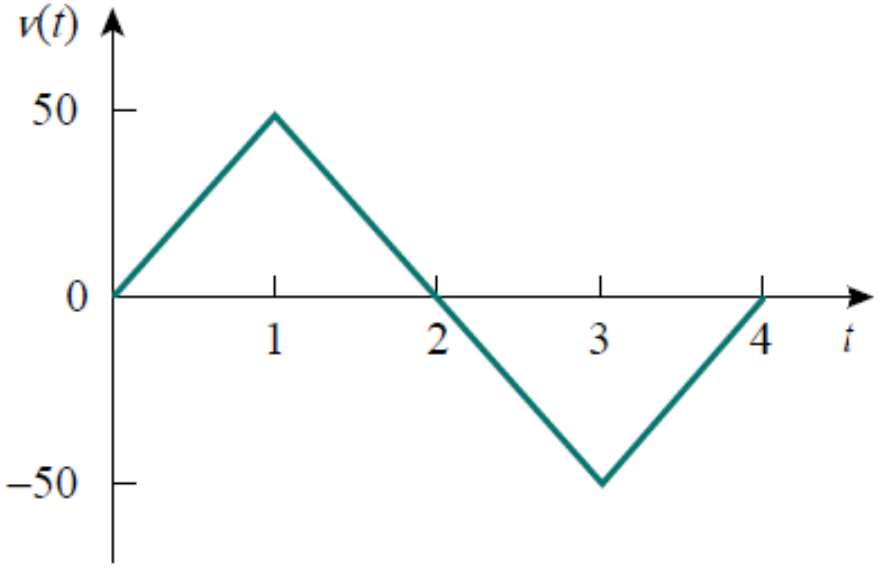
$$i(t) = 6e^{-3000t} \text{ mA}$$

Assume that the initial capacitor voltage is zero.

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EXAMPLE 6.4

Determine the current through a 200-μF capacitor whose voltage is shown in Fig. 6.9.



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PRACTICE PROBLEM 6.4

An initially uncharged 1-mF capacitor has the current shown in Fig. 6.11 across it. Calculate the voltage across it at $t = 2$ ms and $t = 5$ ms.

Answer: 100 mV, 400 mV.

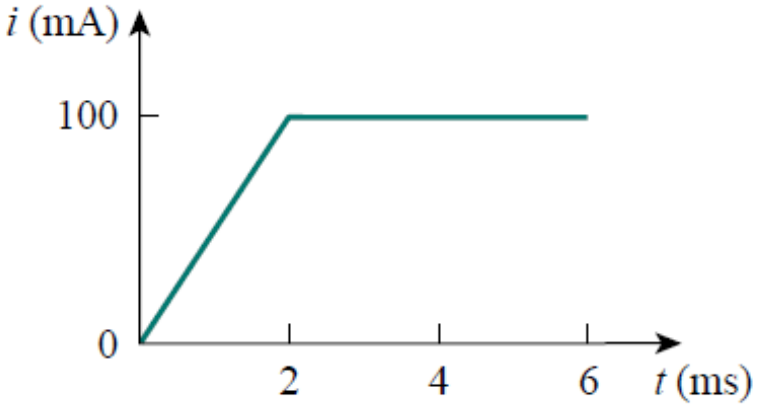


Figure 6.11 For Practice Prob. 6.4.

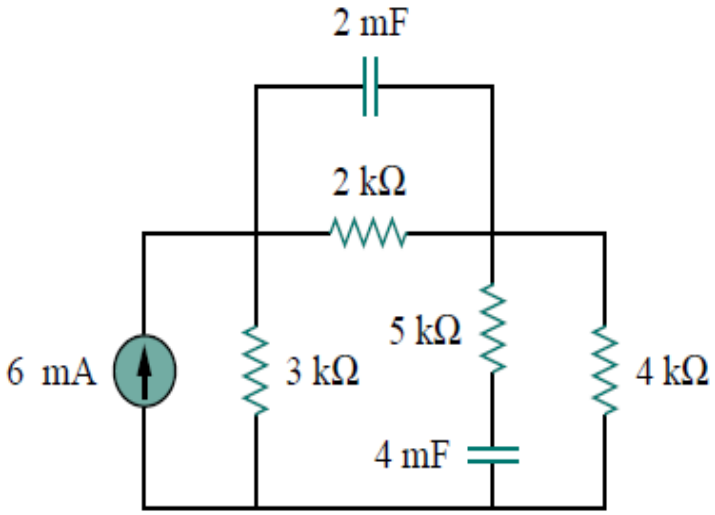
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EXAMPLE 6.5

Obtain the energy stored in each capacitor in Fig. 6.12(a) under dc conditions.



$$w = \frac{1}{2} C v^2$$

Figure 6.12 For Example 6.5.

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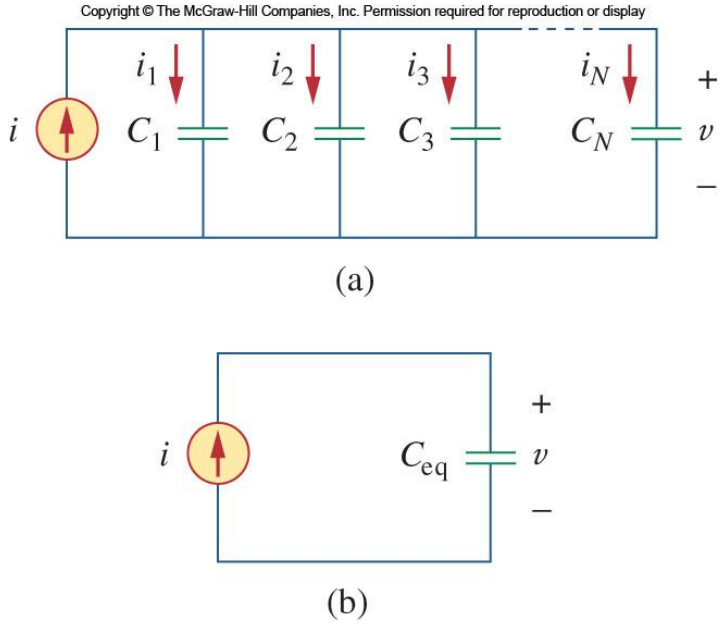
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Parallel Capacitors [1]

- We learned with resistors that applying the equivalent series and parallel combinations can simply many circuits.
- Starting with *N* parallel capacitors, one can note that the voltages on all the caps are the same.
- Applying KCL:

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$$i = i_1 + i_2 + i_3 + \cdots + i_N$$

Parallel Capacitors 2

- Taking into consideration the current voltage relationship of each capacitor:

$$i = C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} + C_3 \frac{dv}{dt} + \cdots + C_N \frac{dv}{dt}$$

$$= \left(\sum_{k=1}^{N} C_k \right) \frac{dv}{dt} = C_{eq} \frac{dv}{dt}$$

- Where

$$C_{eq} = C_1 + C_2 + C_3 + \cdots + C_N$$

- From this we find that parallel capacitors combine as the sum of all capacitance.

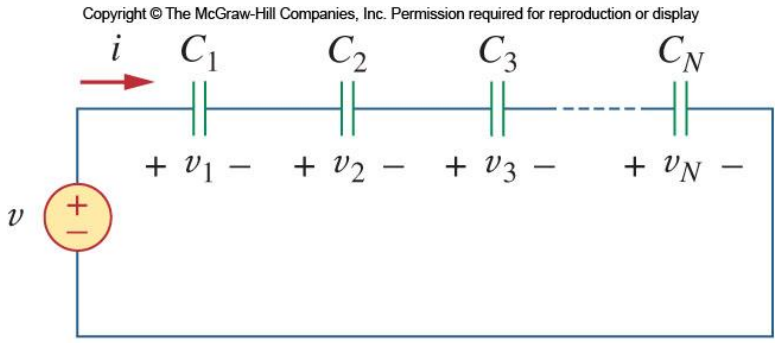
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Series Capacitors [1]

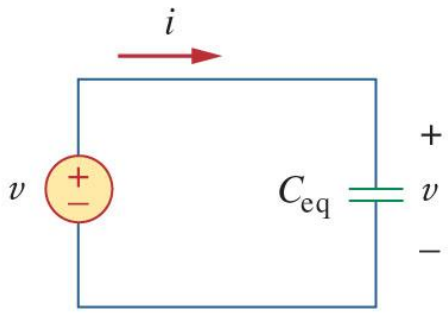
- Turning our attention to a series arrangement of capacitors:
- Here each capacitor shares the same current.
- Applying KVL to the loop:

$$v = v_1 + v_2 + v_3 + \cdots + v_N$$

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(a)



(b)

- Now apply the voltage current relationship.

Series Capacitors 2

$$v = \frac{1}{C_1}\int_{t_0}^{t} i(\tau)\,d\tau + v_1(t_0) + \frac{1}{C_2}\int_{t_0}^{t} i(\tau)\,d\tau + v_2(t_0) + \frac{1}{C_3}\int_{t_0}^{t} i(\tau)\,d\tau + v_3(t_0) + \cdots + \frac{1}{C_N}\int_{t_0}^{t} i(\tau)\,d\tau + v_N(t_0)$$

$$= \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_N}\right)\int_{t_0}^{t} i(\tau)\,d\tau + v_1(t_0) + v_2(t_0) + v_3(t_0) + \cdots + v_N(t_0)$$

$$= \frac{1}{C_{eq}}\int_{t_0}^{t} i(\tau)\,d\tau + v(t_0)$$

- Where

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_N}$$

- From this we see that the series combination of capacitors resembles the parallel combination of resistors.

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The equivalent capacitance of series-connected capacitors is the reciprocal of the sum of the reciprocals of the individual capacitances.

Note that capacitors in series combine in the same manner as resistors in parallel. For $N = 2$ (i.e., two capacitors in series), Eq. (6.16) becomes

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

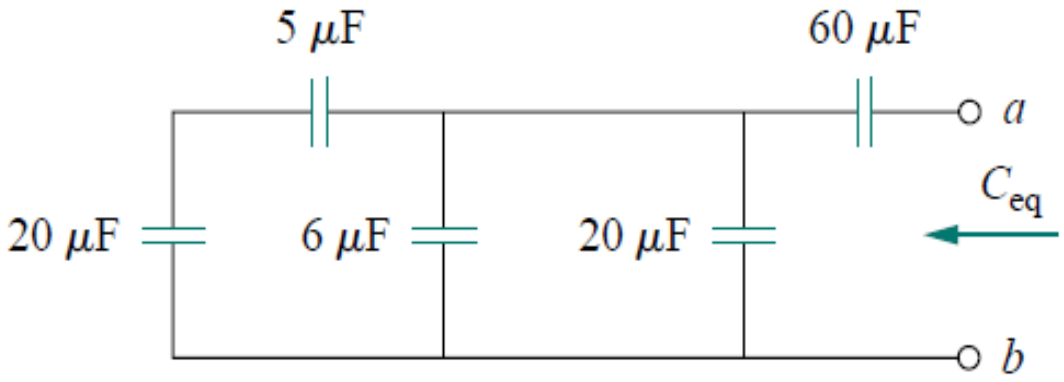
or

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} \tag{6.17}$$

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EXAMPLE 6.6

Find the equivalent capacitance seen between terminals a and b of the circuit in Fig. 6.16.



Ans: 20 μF

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Series and Parallel Caps

- Another way to think about the combinations of capacitors is this:
- Combining capacitors in parallel is equivalent to increasing the surface area of the capacitors:
- This would lead to an increased overall capacitance (as is observed).
- A series combination can be seen as increasing the total plate separation.
- This would result in a decrease in capacitance (as is observed).

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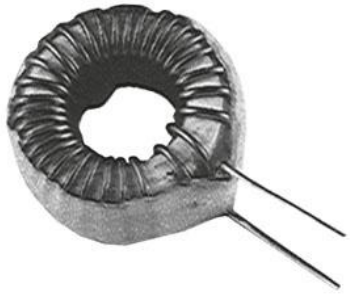
Inductors [1]

- An inductor is a passive element that stores energy in its magnetic field.
- They have applications in power supplies, transformers, radios, TVs, radars, and electric motors.
- Any conductor has inductance, but the effect is typically enhanced by coiling the wire up.

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(a)



(b)



(c)

Courtesy of Tech America

Inductors 2

- If a current is passed through an inductor, the voltage across it is directly proportional to the time rate of change in current.

$$v = L\frac{di}{dt}$$

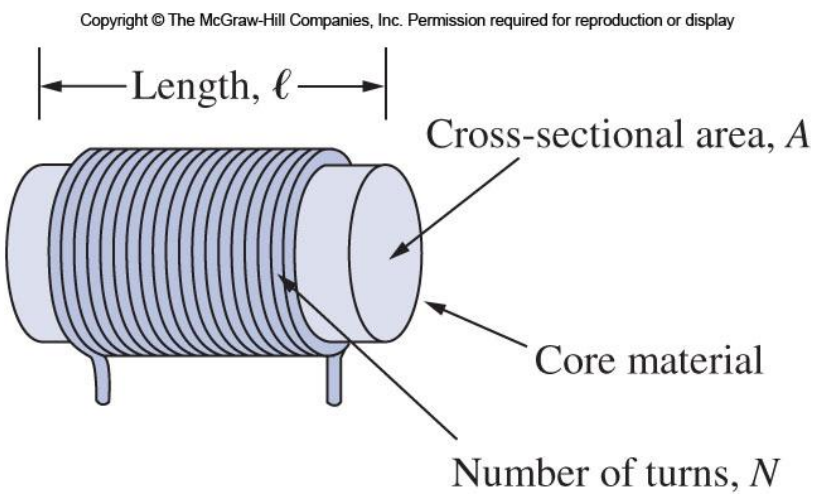
- Where, L, is the unit of inductance, measured in Henries, H.
- On Henry is 1 volt-second per ampere.
- The voltage developed tends to oppose a changing flow of current.

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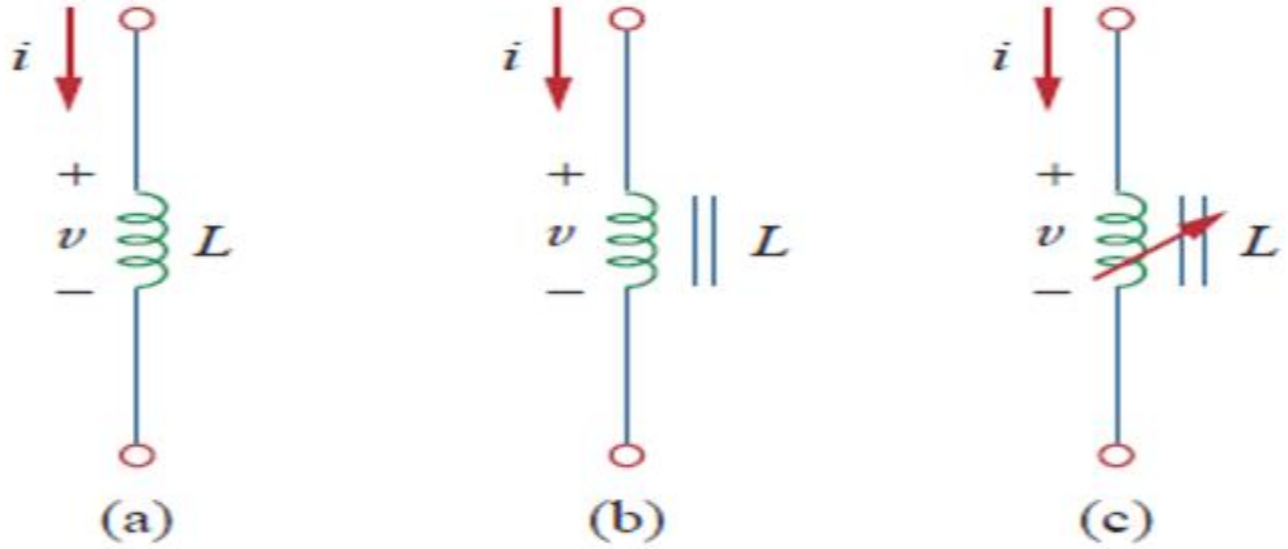
Inductors 3

- Calculating the inductance depends on the geometry:
- For example, for a solenoid the inductance is:

$$L = \frac{N^2 \mu A}{l}$$



- Where N is the number of turns of the wire around the core of cross sectional area A and length *l*.
- The material used for the core has a magnetic property called the permeability, μ.



Circuit symbols for inductors: (a) air-core, (b) iron-core, (c) variable iron-core.

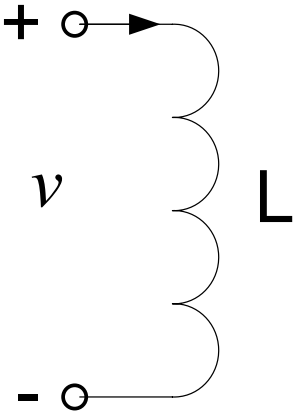
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Current in an Inductor

- The current voltage relationship for an inductor is:

$$I = \frac{1}{L}\int_{t_0}^{t} v(\tau)d\tau + i(t_0)$$

$$v(t) = L\frac{di(t)}{dt}$$



- The power delivered to the inductor is:

$$p = vi = \left(L\frac{di}{dt}\right)i$$

- The energy stored is:

$$w = \frac{1}{2}Li^2$$

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Relationship Between Current and Voltage

$$v(t) = L\frac{di(t)}{dt}$$

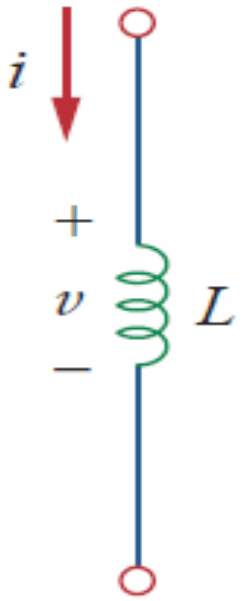
$$di = \frac{1}{L}v\,dt$$

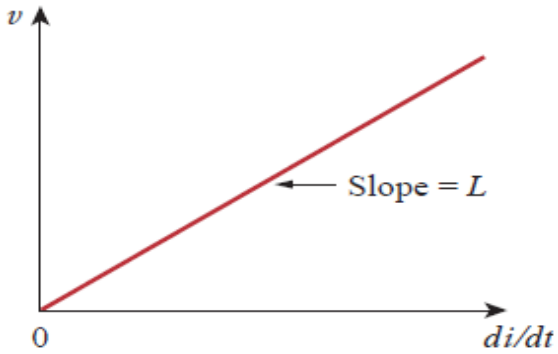
Integrating gives

$$i = \frac{1}{L}\int_{-\infty}^{t} v(t)\,dt$$

Or

$$i = \frac{1}{L}\int_{t_0}^{t} v(t)\,dt + i(t_0)$$





Voltage-current relationship of an inductor.

where $i(t_0)$ = the total current evaluated at t_0 and $i(-\infty) \equiv 0$ (which is reasonable since at some time there was no current in the inductor).

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Properties of Inductors 1

- If the current through an inductor is constant, the voltage across it is zero.
- Thus an inductor acts like a short for DC.
- The current through an inductor cannot change instantaneously.
- If this did happen, the voltage across the inductor would be infinity!
- This is an important consideration if an inductor is to be turned off abruptly; it will produce a high voltage.

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Properties of Inductors 2

- Like the ideal capacitor, the ideal inductor does not dissipate energy stored in it.
- Energy stored will be returned to the circuit later.
- In reality, inductors do have internal resistance due to the wiring used to make them.
- A real inductor thus has a winding resistance in series with it.
- There is also a small winding capacitance due to the closeness of the windings.
- These two characteristics are typically small, though at high frequencies, the capacitance may matter.

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Energy Storage in an Inductor

The instantaneous power delivered to an inductor is

$$p(t) = vi = Li\frac{di}{dt}$$

The energy stored in the magnetic field is thus

$$w_L(t) = \int p(t)dt = L\int_{-\infty}^{t} i\frac{di}{dt}dt = L\int_{-\infty}^{t} idi$$

$$w_L(t) = \frac{1}{2}Li^2(t) \text{ joules}$$

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An inductor has the following important properties:

1. **An inductor acts like a short circuit to dc since from**

$$v(t) = L\frac{di(t)}{dt}$$

$v = 0$ when i = a constant.

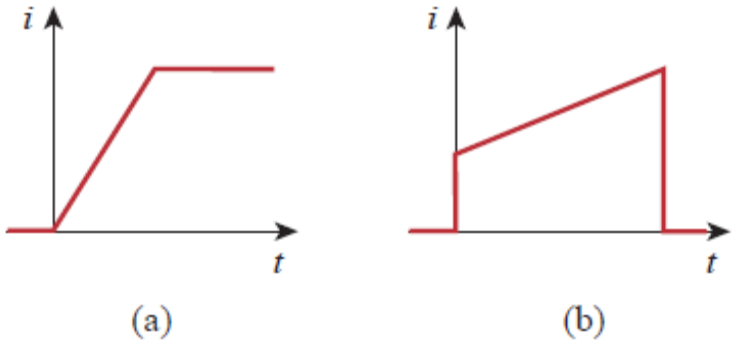


Figure 6.25
Current through an inductor: (a) allowed, (b) not allowable; an abrupt change is not possible.

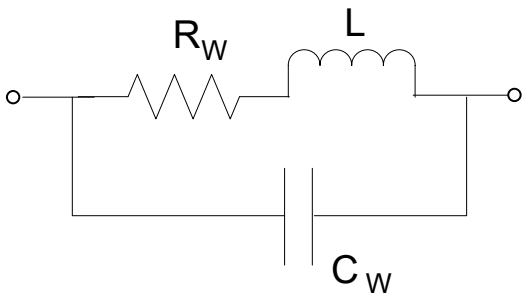
2. **The current through an inductor cannot change instantaneously**, since an instantaneous change in current would require an infinite voltage, which is not physically possible.

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An inductor has the following important properties:

3. Like the ideal capacitor, the ideal inductor does not dissipate energy.

4. A real inductor has a significant resistance due to the resistance of the coil, as well as a "winding capacitance". Thus, the model for a real inductor is shown below.



In this course, however, we will use ideal inductors and assume that an ideal inductor is a good model.

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Example 6.8

The current through a 0.1-H inductor is $i(t) = 10te^{-5t}$ A. Find the voltage across the inductor and the energy stored in it.

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Example 6.9

Find the current through a 5-H inductor if the voltage across it is

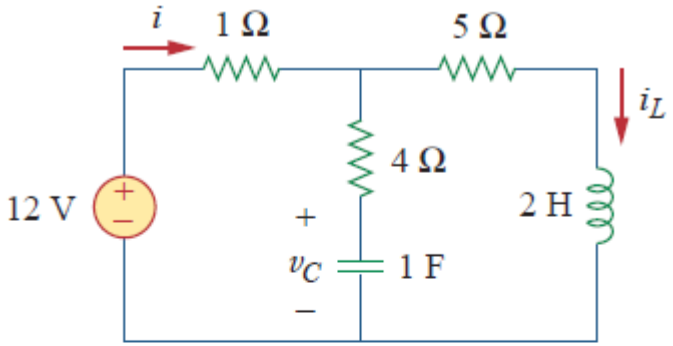
$$v(t) = \begin{cases} 30t^2, & t > 0 \\ 0, & t < 0 \end{cases}$$

Also, find the energy stored at $t = 5$ s. Assume $i(v) > 0$.

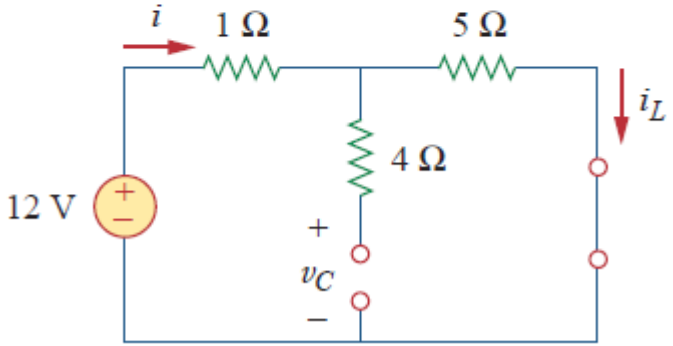
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Example 6.10

Consider the circuit in Fig. 6.27(a). Under dc conditions, find: (a) i, v_C, and i_L, (b) the energy stored in the capacitor and inductor.



(a)



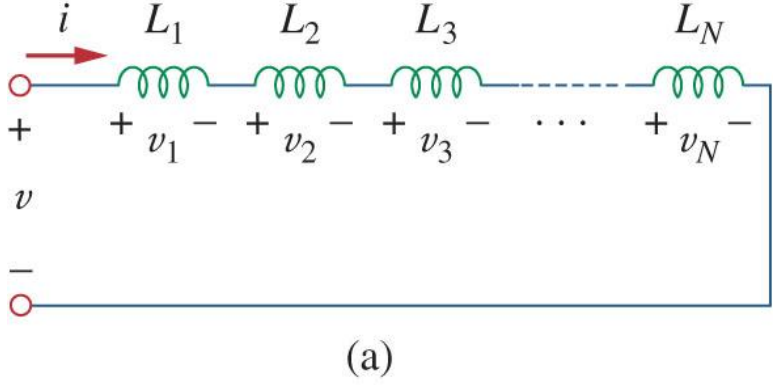
(b)

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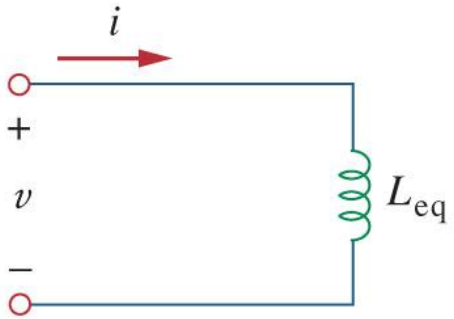
Series Inductors [1]

- We now need to extend the series parallel combinations to inductors.
- First, let's consider a series combination of inductors.
- Applying KVL to the loop:

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(a)



(b)

$$v = v_1 + v_2 + v_3 + \cdots + v_N$$

Series Inductors 2

- Factoring in the voltage current relationship.

$$v = L_1 \frac{di}{dt} + L_2 \frac{di}{dt} + L_3 \frac{di}{dt} + \cdots + L_N \frac{di}{dt}$$

$$= \left(\sum_{k=1}^{N} L_k \right) \frac{di}{dt} = L_{eq} \frac{di}{dt}$$

- Where

$$L_{eq} = L_1 + L_2 + L_3 + \cdots + L_N$$

- Here we can see that the inductors have the same behavior as resistors.

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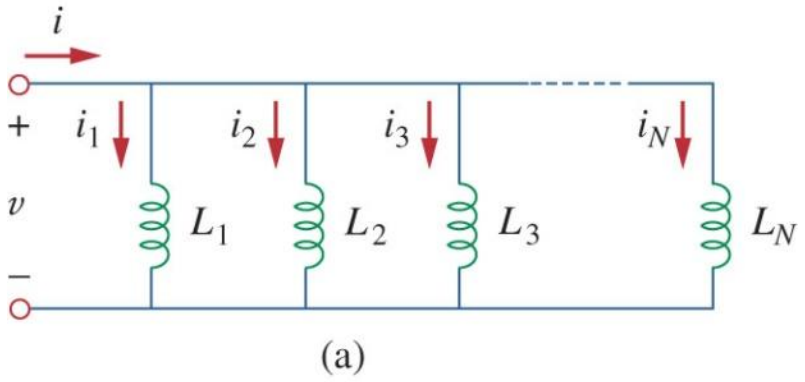
Parallel Inductors [1]

- Now consider a parallel combination of inductors:
- Applying K C L to the circuit:

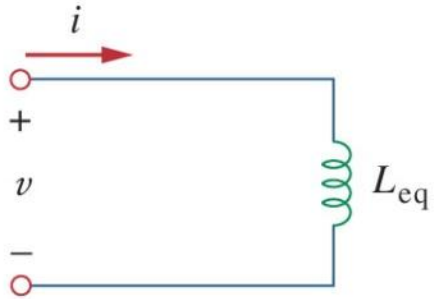
$$i = i_1 + i_2 + i_3 + \cdots + i_N$$

- When the current voltage relationship is considered, we have:

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(a)



(b)

$$i = \left(\sum_{k=1}^{N} \frac{1}{L_k} \right) \int_{t_0}^{t} v\,dt + \sum_{k=1}^{N} i_k(t_0) = \frac{1}{L_{eq}} \int_{t_0}^{t} v\,dt + i(t_0)$$

Parallel Inductors 2

- The equivalent inductance is thus:

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \cdots + \frac{1}{L_N}$$

- Once again, the parallel combination resembles that of resistors.

The **equivalent inductance** of parallel inductors is the reciprocal of the sum of the reciprocals of the individual inductances.

Note that the inductors in parallel are combined in the same way as resistors in parallel.

For two inductors in parallel ($N = 2$), Eq. (6.30) becomes

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} \quad \text{or} \quad L_{eq} = \frac{L_1 L_2}{L_1 + L_2} \tag{6.31}$$

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Summary of Capacitors and Inductors

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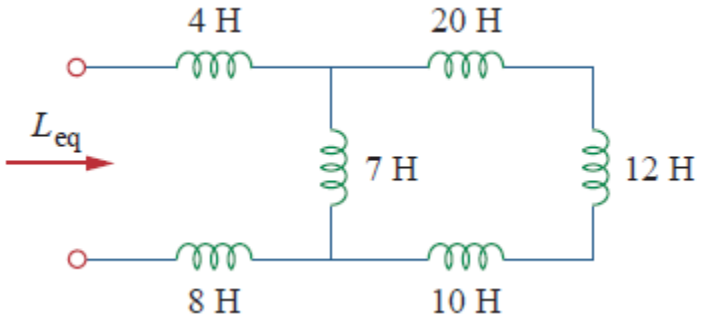
TABLE 6.1

Important characteristics of the basic elements.†

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
v-i:	$v = iR$	$v = \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau + v(t_0)$	$v = L\frac{di}{dt}$
i-v:	$i = v/R$	$i = C\frac{dv}{dt}$	$i = \frac{1}{L}\int_{t_0}^{t} v(\tau)d\tau + i(t_0)$
p or w:	$p = i^2R = \frac{v^2}{R}$	$w = \frac{1}{2}Cv^2$	$w = \frac{1}{2}Li^2$
Series:	$R_{eq} = R_1 + R_2$	$C_{eq} = \frac{C_1C_2}{C_1 + C_2}$	$L_{eq} = L_1 + L_2$
Parallel:	$R_{eq} = \frac{R_1R_2}{R_1 + R_2}$	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit
Circuit variable that cannot change abruptly:	Not applicable	v	i

Example 6.11

Find the equivalent inductance of the circuit shown in Fig.



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Example 6.12

For the circuit in Fig. 6.33, $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find: (a) $i_1(0)$; (b) $v(t)$, $v_1(t)$, and $v_2(t)$; (c) $i_1(t)$ and $i_2(t)$.

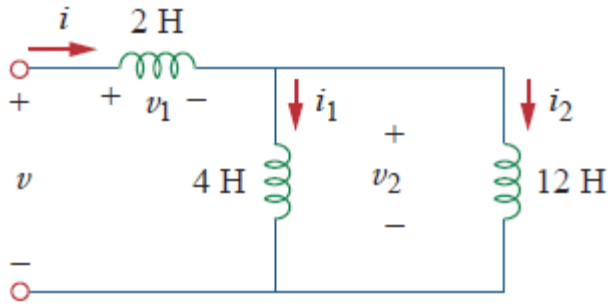


Figure 6.33
For Example 6.12.

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Example 6.12

For the circuit in Fig. 6.33, $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find: (a) $i_1(0)$; (b) $v(t)$, $v_1(t)$, and $v_2(t)$; (c) $i_1(t)$ and $i_2(t)$.

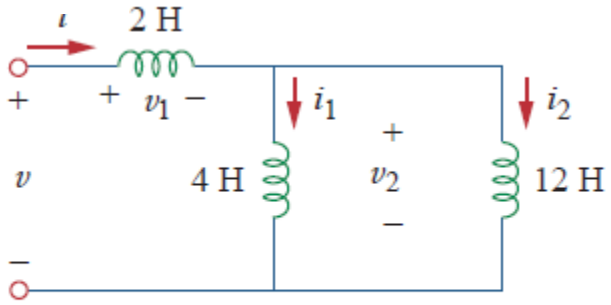


Figure 6.33
For Example 6.12.

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